

GABRIEL NAVARRO CERÓN

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EDUCATION

INSTITUTO TECNOLÓGICO AUTÓNOMO DE MÉXICO (ITAM)

Bachelor of Science in Data Science

Mexico City, Mexico

Expected June 2028

Bachelor of Science in Computer Engineering

Expected June 2028

- GPA: **97.3 / 100** | Granted **ITAM Merit Scholarship (75%)**
- **Academic Excellence Award** — Highest GPA in Computer Engineering (2025).
- *Relevant Coursework:* **Artificial Intelligence**, Data Engineering & Distributed Systems, Probability Theory, Advanced Data Structures & Algorithms, Reinforcement Learning, Software Engineering, Database Systems & NoSQL, Linear Algebra.

MODERN AMERICAN SCHOOL

High School Diploma

Mexico City, Mexico

Graduated June 2023

- GPA: 96 / 100 | **UNAM-SI Full-ride Scholarship (100%)** by D-GIRE UNAM - National merit award for academic excellence.

EXPERIENCE

DATA, ALGORITHMS, AND SOCIETY CENTER (CDAS) ITAM

Software Engineer Intern

PROJECT FOR SEARCH COLLECTIVES

Mexico City, Mexico

August 2025 - Current

- Architected nation-scale missing persons search platform centralizing **10,000+ records** from **40+ sources** via Python, FastAPI, PostgreSQL, and React, deployed on **AWS ECS with Fargate**, aggregating life-critical data for families and search collectives.
- Engineered a hybrid **Levenshtein fuzzy-matching** algorithm optimized for Spanish naming patterns and integrated **AWS Rekognition** photo-matching, delivering **<120ms** query responses and **95%+** recall on incomplete queries.
- Built ingestion pipeline processing HTML, PDF, and image files at 120 records/min via parallelization and fault-tolerant design.
- Designed a distributed scheduler with **dynamic interval tuning**, exponential backoff, and **shard-based partitioning** per scraper, achieving **99% extraction success rate** across 40+ sources via probabilistic reliability modeling.

ITAM CODING RUSH | *Competitive Programming Competition & Laboratory*

Organizer & Course Developer

Mexico City, Mexico

August 2024 - Current

- Organized olympiad-level programming competition for **100+ students**, designing a curriculum and **20+ problems** spanning graph theory, dynamic programming, and combinatorial optimization; co-founded weekly algorithm lab for 15+ university students.

PROJECTS

FAULTMAP | *Open-Source, Pip-Installable LLM Diagnostic & Evaluation Library*

Python, HuggingFace, PyTorch, HDBSCAN, Scikit-Learn, NumPy, asyncio, Docker, Pytest, Github Actions (CI/CD)

- Engineered an **LLM failure-slice detection pipeline**, embedding prompts via **HuggingFace** and clustering semantically similar inputs with **HDBSCAN** to run per-cluster hypothesis tests and surface statistically elevated failure slices, powering a parallel **coverage auditor** that automates the LLM-naming of semantic blind spots in live production traffic.
- Designed autonomous scoring via **semantic entropy** and multi-sample self-consistency, measuring embedding-space divergence across multiple LLM responses per prompt at high temperature to **flag hallucination-prone regions** without ground-truth labels.
- Implemented native Python statistical engine with **McNemar's** test for model comparison, **Benjamini-Hochberg FDR** correction across all hypothesis tests, and automatic **Fisher exact/chi-squared** routing by sample size – no scipy or statsmodels dependencies.
- Deployed a production-grade async pipeline handling **50+ parallel LLM** calls with semaphore-based rate limiting and exponential backoff; **published to PyPI** via GitHub Actions **CI/CD** with an interactive Colab tutorial and Apache 2.0 license.

OPTIONS FLOW IMPACT MONITOR | *Python, PySpark, LightGBM, Apache Kafka, FastAPI, MongoDB, Parquet, React, Docker*

- Architected **9-container MongoDB sharded cluster** (2 replica-set shards, hashed sharding on option symbols, RBAC) capturing live options microstructure data via parallel WebSocket consumer and 60-second REST snapshot poller.
- Built **3-stage PySpark** batch pipeline (clean, enrich, transform) computing **5 institutional analytics** - Vol/OI ratio, IV Skew, Net Gamma Exposure, Put/Call ratio, and spread vs. moneyness - across **10 live tickers**.
- Deployed **LightGBM** volatility forecasting microservice via FastAPI with online/offline decoupled **MLOps** architecture; enforced temporal correctness via **point-in-time joins** on a **Parquet**-based feature store to eliminate look-ahead bias.

ALGORITHMIC WORDLE OPTIMIZATION ENGINE | *1st Place Tournament Winner* | *Python, NumPy, Multiprocessing*

- Achieved a **>99% solve rate** at **~3.7 average guesses** under strict 1-CPU/5-second latency constraints via Shannon entropy maximization and dynamic probability scoring over the full 27^N combinatorial space.
- Accelerated state-space search by **~640x** via **NumPy vectorization** and multiprocessing, precomputing optimal decision trees for turns 1 to 3 via serialized lookup tables with **zero I/O overhead** at inference time.
- Engineered cluster-busting heuristics resolving high-collision states via non-dictionary probes that maximized partition entropy.

ML TRAFFIC CRASH RECIDIVISM PREDICTION | *Python, PostgreSQL, XGBoost, LightGBM, Scikit-learn, Geopandas*

- Architected an ML pipeline for **8M+ records** with SQL feature marts and **temporal cross-validation** for **leakage-free forecasting**.
- Designed an XGBoost classification stack, engineering **40+ predictive features**, including DBSCAN and K-means for geospatial hotspot scores achieving **46% recall** in identifying the top 5% highest-risk zones while managing high class imbalance.

TECHNICAL SKILLS

- **Languages:** Python, Java, C++, SQL, R, Bash/Shell, MATLAB, TypeScript
- **ML & AI:** PyTorch, HuggingFace (sentence-transformers), Scikit-learn, LightGBM, XGBoost, HDBSCAN, NumPy, SciPy
- **Data Engineering:** Polars, Pandas, PostgreSQL, MongoDB, Cassandra, Neo4j, Apache Spark (PySpark), Apache Kafka, Parquet
- **Infrastructure:** Git/GitHub, Docker, FastAPI, React, AWS (ECS, Fargate, Rekognition), GitHub Actions (CI/CD), Linux/WSL

AWARDS & HONORS

- Mexican Math Olympiad – National Silver (2023), National Bronze (2022).
- Mexican Chemistry Olympiad – National Silver (2023) | Gauss Mathematics Contests – 2nd Place National (2021).
- **Languages:** Spanish (Native), English (C2 Proficiency, CPE), French | **Interests:** basketball, trekking, chess, music, travelling.